

Chronic Kidney Disease (CKD) Prediction using Machine Learning

1. Problem Statement

The objective of this project is to develop a predictive model for identifying **Chronic Kidney Disease (CKD)** based on various patient health parameters. Since the target variable (**classification**) is categorical (Yes/No), this problem is a **Supervised Machine Learning Classification** task.

2. Dataset Information

The dataset consists of **399 records (rows)** and **25 attributes (columns)**. The target variable is **classification**, which indicates whether a patient has Chronic Kidney Disease or not.

For model development, the **classification** column was separated as the target (output) variable, while the remaining 24 columns were used as the input features.

3. Data Pre-processing

During the data preprocessing stage, all categorical (string) variables were converted into numerical values using **One-Hot Encoding (Dummy Variables)**, as these features represent nominal categorical data.

The preprocessing steps included:

- Separating the input and output variables.
- Applying One-Hot Encoding to convert categorical variables into binary (0 and 1) values.
- Preparing the dataset for machine learning model training.

4. Model Development

A **Support Vector Machine (SVM)** classifier was developed to predict Chronic Kidney Disease. To improve the model's performance, **GridSearchCV** was applied to identify the optimal hyperparameter values.

The final model achieved the following performance:

- **Accuracy:** 99%
- **Weighted F1-Score:** 99%

Overall Performance

- **Accuracy:** 99%
- **Macro Average F1-Score:** 99%
- **Weighted Average F1-Score:** 99%

5. Model Evaluation

The performance of the proposed model was evaluated using multiple classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

The model achieved a **ROC-AUC score of 1.00**, indicating excellent classification performance and a high ability to distinguish between CKD and non-CKD patients.

6. Final Model Selection

The Support Vector Machine (SVM) classifier was selected as the final model because it demonstrated excellent predictive performance. After hyperparameter tuning using GridSearchCV, the model achieved:

99% Accuracy

99% Weighted F1-Score

ROC-AUC Score of 1.00

These evaluation results indicate that the SVM model provides highly accurate and reliable predictions for Chronic Kidney Disease classification, making it the most suitable model for this dataset.